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Deep eutectic solvents boosting solubilization and Se-functionalization of heteropolysaccharide: Multiple hydrogen bonds modulation

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ABSTRACT

In this study, the favorable feasibility of deep eutectic solvents (DESs) in solubilization and functionalization of natural heteropolysaccharide was validated by experiments and density functional theory calculations. This revealed that choline chloride-based DES/DMSO (dimethyl sulfoxide) binary mixed solvents possessed more and stronger hydrogen bonding sites, facilitating the balance between disruption and reconstruction of hydrogen bonds within branched heteropolysaccharide from *Artemisia sphaerocephala* (PAS) and achieving efficient solubilization. Further, due to the full exposure and activation of polysaccharide hydroxyls, the efficiency of DES/DMSO-mediated novel Se-functionalization was substantially enhanced compared to the conventional selenylation methods. The derivative exhibited conversion to lower molecular mass with rigid solution conformation based on co-solvent effect and predominant acidic environment influence. This study offered a framework for exploring the potential of individualized polysaccharide functionalization by modulating DES constituents to achieve multiple controllabilities in terms of conversion efficiency and derivative structure.

1. Introduction

The ever increasing demands for resources in modern society and the promotion of environmental protection have led to a growing global concern for sustainable natural biological resource (Xu, Che, et al., 2019). In particular, polysaccharide resources of natural biological origin have been used in fuels, materials and pharmaceuticals (Mika et al., 2018). Limitations of structural properties make many natural polysaccharides biologically inactive, hence functionalization strategy becomes an effective way to develop polysaccharide resources (Li et al., 2016). However, owing to the complex structures and the presence of multiple hydrogen bond networks of polysaccharides, organic solvents that mediate polysaccharides functionalization in traditional reaction systems present poor abilities to dissolve polysaccharides and drawbacks such as wastewater contamination and biological toxicity (Farrán et al., 2015). The resulting non-homogeneous environment generally leads to reduced functionalization efficiencies of polysaccharides, which is not conducive to the exploitation of biological resources. Accordingly, the functionalized valorized development of natural resources requires new solvents and reaction media to achieve the mutual promotion and coordination between the efficient utilization of resources and ecological environmental protection.

Green solvents are a class of environmentally friendly solvents that have been widely used in the development of biological resources in recent years, among which deep eutectic solvents (DESs) prepared by the hydrogen bond interactions between hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs) are considered to possess great potential for macromolecular biotransformation (Hansen et al., 2020; Zhang et al., 2012). The hydrogen bond providing/accepting ability of DESs can break hydrogen bonds between carbohydrates, while DESs exhibit high designability, low toxicity and low volatility (Chen & Mu, 2019). At the same time, the high viscosity and thermal decomposition exhibited by parts of DESs (González-Rivera et al., 2020; Rodriguez Rodriguez et al., 2019) make the application of DESs to the processing of natural polysaccharides full of opportunities and challenges. Moreover, the great dissolving performance of DES for polysaccharide could be enhanced by building binary mixed solvents system to further reduce the viscosity, leading to lower experimental cost and

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effective substrate conversion. The dual solvent-solute/solvent interaction existing in the mixed solvents can further affect the reaction efficiency via solvent effects (Reichardt & Welton, 2011; Ruan et al., 2021). Hereby, much attention has been paid to investigating the interaction between co-solvents and DESs as well as the effect of modulating solvents on the functionalization of polysaccharide resources.

Se-functionalization is an important approach to enhance the activity of natural polysaccharides and has been widely reported in recent years as a promising modification of natural polysaccharides (Li et al., 2019; Xu, Wu, et al., 2019). Our previous works achieved selenylation of natural heteropolysaccharide and revealed that in case of modified heteropolysaccharide, the introduction of anion functional groups and the tendency to a rigid solution conformation improved the bioactivities of anti-tumor (Liu et al., 2021) and immunomodulatory (Li et al., 2020). Moreover, compared with homopolysaccharide, heteropolysaccharide with diverse monosaccharide components and more branches resulting in critical problem of low substrate solubility, thereby leading to low subsequent conversion efficiency and harsh reaction environment. This is particularly evident in conventional polysaccharide selenylation systems, thus making the solvent as a key entry point for the construction of novel polysaccharide selenylation system (Guo et al., 2017). It has been demonstrated in many works that the hydrogen bond property of DES contributes to providing more hydrogen bonding sites, which is beneficial to break the hydrogen bond networks and promote the solubilization and subsequent functionalization reactions of polysaccharide (Cao et al., 2019). However, there is a relative paucity of empirical research specifically addressing the feasibility of DES for dissolving heteropolysaccharide and the effect of solvent composition regulation on dissolution capacity. For the highly solvent-dependent heteropolysaccharide processing, it is necessary to explore the prospect of DES as an effective solvent and reaction medium in the rather difficult selenylation reaction and expand the applicability of DES in the field of polysaccharide functionalization.

Artemisia sphaerocephala is a kind of sand-fixing plant widely distributed in northwest China, and a branched heteropolysaccharide extracted from its seeds (PAS) showed a weight average molecular mass (M_W) of 6.96 × 10^4 g/mol (Wei et al., 2019). In this work, in order to break the bottleneck caused by heteropolysaccharide solubility limitation, a DES-mediated solvent system for PAS was proposed, and further quantum chemical calculations were used to investigate the interactions within mono and binary solvents systems to speculate the possible solubility mechanism. Then, efficient DES-mediated Se-functionalizations of PAS and corresponding structural confirmations were carried out to explore the effect of this selenylation system on the conversion efficiency and heteropolysaccharide chain features. It is hoped that this research will contribute to a deeper understanding regarding the effect of DES on heteropolysaccharide solubilization and provide a reference for the investigation of DES performance modulation in biomolecular functionalization.

2. Materials and methods

2.1. Materials

Dimethyl sulfoxide (DMSO), choline chloride (ChCl), tetramethylammonium chloride (TMAC), ethylene glycol (EG), urea, glycerin (GLY), 1,4-butanediol (BDO), trifluoroacetic acid, hydroxylamine hydrochloride, pyridine, chloroform, Reichardt's dye (30) (RD), *N,N*-dimethyl-4-nitroaniline (DMNA), 4-nitroaniline (NA), ascorbic acid, potassium bromide (KBr), 2,3-diaminonaphthalene (DAN), ethylenediamine tetraacetate (EDTA-2Na), congo red (CR) and methylene blue (MB) were purchased from Jingchun Industry Co., Ltd., Shanghai, China. All chemicals used in present work were analytical reagents or better grades with no further purification.

PAS was extracted and purified according to our previous study (Wei et al., 2019). In brief, crude PAS was extracted with distilled water at 80 °C for 18 h. All extraction solution was concentrated and then deproteinated using Sevag method to obtain purified PAS with M_W of 6.96 × 10^4 g/mol. The monosaccharide compositions of PAS were L-arabinose (Ara), D-xylose (Xyl), D-mannose (Man), D-glucose (Glc) and D-galactose (Gal) in molar ratio of 1: 4.33: 3.21: 4.63: 2.57.

2.2. Preparation of DESs

DESs were prepared by mixing the HBAs with the corresponding HBDs according to the molar ratios provided in Table 1, and all the relevant materials had been dried prior to preparation. The mixture in the sealed flask was stirred with a magnetic stirrer and heated at 80 °C until clear homogenous liquid was obtained. Heated for another 2 h, the mixture was placed in a glass vial and kept in a vacuum chamber at 50 °C to remove moisture from DES, then stored the glass vial in a desiccator to prevent moisture absorption.

2.3. Polarity determination of DESs

The polarity scale (E_T^N), dipolarity/polarizability (π^*), hydrogen bond acidity (α) and hydrogen bond basicity (β) were calculated according to modified Kamlet-Taft empirical polarity scales (Ranjkesh et al., 2018).

$$E_T(30)/\text{kcal}\cdot\text{mol}^{-1} = 28591/\lambda_{max(\text{RD})} \quad (1)$$

$$E_T^N/\text{kJ}\cdot\text{mol}^{-1} = (E_T(30) - 30.7)/32.4 \quad (2)$$

$$\pi^* = (28.18 - \upsilon_{(\text{DMNA})})/3.52 \quad (3)$$

$$\beta = (0.9841 \times \upsilon_{(\text{DMNA})} + 3.49 - \upsilon_{(\text{NA})})/2.759 \quad (4)$$

$$\alpha = (E_T(30) - 31.2 - 11.5\pi^*)/15.2 \quad (5)$$

where,

$$\upsilon_{(\text{NA})} = 1/(\lambda_{max(\text{NA})} \times 10^{-4})\text{kK} \quad (6)$$

$$\upsilon_{(\text{DMNA})} = 1/(\lambda_{max(\text{DMNA})} \times 10^{-4})\text{kK} \quad (7)$$

Here, $\lambda_{\text{max(RD)}}$ [nm] represented the maximum absorption wavelength of Reichardt's dye (30). $\lambda_{\text{max(NA)}}$ [nm] and $\lambda_{\text{max(DMNA)}}$ [nm] represented the maximum absorption wavelength of 4-nitroaniline and *N, N*-dimethyl-4-nitroaniline.

2.4. Measurement of viscosity

The viscosity was measured using Physica MCR 301 rheometer (Anton Paar, Austria) with a 50 mm parallel plate (PP50). Solvent viscosity measurement was performed at temperatures ranging from 20 °C to 95 °C at 10 °C intervals. Shear-dependence viscosity of solvent was measured at 25 °C and 80 °C in a shear rate range between 0.1 and 100 s^{-1}.

Table 1
Polarities and Kamlet-Taft parameters of the prepared DESs.

DES	HBA	HBD	HBA/HBD molar ratio	E_T^N	π^*	α	β
DES_1	ChCl	Urea	1:2	0.814	1.193	0.799	0.453
DES_2	ChCl	EG	1:2	0.824	1.093	0.897	0.554
DES_3	ChCl	GLY	1:2	0.839	1.127	0.902	0.487
DES_4	ChCl	BDO	1:2	0.793	1.026	0.881	0.614
DES_5	TMAC	Urea	1:2	0.773	1.110	0.774	0.508
DES_6	TMAC	EG	1:2	0.783	0.991	0.886	0.633
DES_7	TMAC	GLY	1:2	0.790	1.008	0.887	0.512
DES_8	TMAC	BDO	1:2	0.766	0.974	0.863	0.680

2.5. Solubility of PAS in different solvents

The detailed measurement procedure was according to a recent study (Cao et al., 2019). Briefly, PAS was accurately weighed (5 g ± 0.1 mg) into different solvents (10 g ± 0.1 mg) and the mixture was magnetically stirred at 80 °C for 5 h. The dissolved samples were centrifuged at 10000 rpm for 30 min, and the supernatant was collected. The supernatant was diluted 50 times with deionised water, then 1 mL of deionised water and 1 mL of 6% (w/v) phenol were added and mixed well. Next, 5 mL of concentrated sulfuric acid was added to the mixture and mixed well. After standing for 30 min, the absorption spectrum of the mixture at 490 nm was measured and calculated according to the following equation:

$$S = (A_{490nm} - b)/a \times 0.9 \times 10 \quad (8)$$

where S represented the solubility of PAS (g/100 g), the absorbance of the mixture at 490 nm was expressed as A_{490nm}, the slope and intercept of the standard curve were expressed as a and b. The standard curve was obtained by dissolving 0.1, 0.2, 0.3, 0.4 and 0.5 g of glucose into 10 g of different DESs according to the above procedure.

The determination process was performed in triplicate. The experimental data were analyzed using SPSS V.17.0 ($p < 0.05$).

2.6. Polarized light microscope

0.25 g PAS was put into 50 g solvent and stirred at 80 °C for 5 h. One drop of solution was transferred onto the slide and covered with coverslip. The polarized light microscopy (Axio scope A1, Carl Zeiss, Germany) was performed to observe PAS granules and capture images.

2.7. Computational methods

The initial structure optimization of DES was performed at HF/6-311G(d,p) level, and further reoptimization was obtained using the M06-2X exchange-correlation function in combination the 6-31++G(d,p) basis set based on the Gaussian 16 program package (Zhao & Truhlar, 2008). In order to reveal the non-covalent interactions in DES and DES/DMSO systems, the reduced density gradients (RDG) was calculated using Multiwfn program and the data were visualized by VMD and Gnuplot (Lu & Chen, 2012a). The electrostatic surface potential (ESP) of molecule was used to analyze hydrogen bond interactions, and the locations of the surface extrema and the statistical analysis of van der Waals (vdW) surface area in different ESP ranges were calculated referring to previous works (Lu & Chen, 2012b; Lu & Manzetti, 2014). For atoms in molecules (AIM) analysis, the electron density and Laplace amount of electron density at the bond critical point (BCP), namely ρ(BCP) and $\nabla^2\rho$(BCP), were commonly used to analyze the bond strength as well as covalent character (Lu & Chen, 2013). The reactivity of the solvent system was further analyzed by frontline molecular orbitals (FMO) theory in combination with Multiwfn program and VMD.

2.8. Se-functionalization of PAS

The preparation of Se-functionalized PAS (SePAS) was achieved according to the previously reported method (Tao et al., 2021; Xiao et al., 2019) with some modifications. To ensure the same reaction environment volume, the pure DES reaction volume was 60 mL, and the mixed solvents consisted of 30 mL DES and 30 mL DMSO. The above solvents were separately placed in three neck vials. PAS (300 mg) was added into the vial at 80 °C while stirring in oil bath for 5 h. Next, H_2SeO_3 (1.25 g) was added and stirred at 80 °C for 5 h under nitrogen atmosphere. After the reaction was complete, the sample was centrifuged at 10000 rpm for 10 min. The collected supernatant was dialyzed with distilled water for 48 h (molecular mass cut-off 5–8 kDa) until the solution was colorless when ascorbic acid was added. Finally, the mixture was centrifuged at 8000 rpm for 5 min. The collected supernatant was evaporated, precipitated with acetone and lyophilized for 48 h to obtain SePAS.

2.9. FT-IR spectra

The powdery samples were prepared by grinding mixture of 1 mg of sample with 200 mg of dried KBr and pressing into slices. The liquid samples were prepared by liquid films method using KBr slices. The spectra were recorded at the wavenumber range of 4000–400 cm^{-1} with FT-IR spectrophotometer (iS10, Thermo Nicolet, USA). The number of scans was 16 and referenced against air. The resolution of FT-IR was 4 cm^{-1}.

2.10. NMR analysis

^1H, ^{13}C, $^1H-^1H$ COSY and ^1H-^{13}C HSQC NMR spectra were recorded at 25 °C by a Bruker Avance DPX-400 spectrometer (Bruker, Rheinstetten, Germany). Chemical shifts were reported in ppm using TMS as the internal reference. The NMR spectra of DESs and polysaccharides were determined in DMSO-d_6 and D_2O, respectively.

2.11. Determination of selenium content

The selenium contents of SePASs were determined as described in our previous study (Wei et al., 2019). The mixture of 2 mL of acid (H_2SO_4 : HNO_3 : $HClO_4$=1 : 1:4, V/V) was used to hydrolyze the sample in microwave digestion system (Multiwave 3000, Anton Paar, Austria) for 1.5 h. After cooling, the digestion solution was concentrated to 2 mL and 8 mL of 0.1 mol/L hydrochloric acid was added. To the mixed solution, added 4 mL of 0.5 mol/L hydroxylamine hydrochloride and 0.2 mol/L EDTA-2Na, mixing well and standing for 10 min. Then added 2 mL of 0.1% DAN solution and placed it in dark place for storage. The mixture was boiled in water bath for 5 min and then cooled and extracted with 5 mL of cyclohexane. Fluorescence spectrometry (RF-5301PC, Shimadzu, Japan) was used for detecting the fluorescence intensity of cyclohexane solution (excitation and emission wavelength at 376 and 520 nm, respectively). The selenium content in the sample was calculated from the standard curve.

2.12. Molecular mass analysis

The molecular mass of the sample (1 mg/mL) was determined via size-exclusion chromatography combined with multi-angle laser light scattering (SEC-MALLS). The sample was filtered through a 0.45 μm cellulose filter and eluted by a Ultrahydrogel TM column (7.8 × 300 mm, Waters, USA) with ultrapure water as mobile phase. The injection volume and flow rate were 50 μL and 0.5 mL/min, respectively. The refractive index increment (dn/dc) value was determined to be 0.145 mL/g. *Zimm* method was used to calculate the M_W and data analysis was performed using Astra software (Wyatt Tech. Corp., USA).

2.13. SePAS/dye interaction analysis

2 mL of each polysaccharide solution with different concentrations (0.01–0.2 mg/mL) was added to 2 mL of MB solution (0.02 mg/mL) at room temperature respectively. After shaking at 70 °C for 10 min, the solution was warmed up to 90 °C and then shaken for 20 min in water bath. The absorption spectra were performed after cooling (UV1000, Labtech, China). 2 mL of polysaccharide solution (0.02–0.5 mg/mL) was mixed well with 2 mL of CR dye (91 μg/mL) at room temperature. In addition, 2 mL of NaOH solution (0.1–2 mg/mL) was added to 4 mL of polysaccharide (0.2 mg/mL)/CR mixed solution, and the absorption spectra of the above samples were recorded in the range of 300–700 nm (UV1000, Labtech, China).

2.14. Monosaccharide composition analysis

The sample (10 mg) was hydrolyzed with trifluoroacetic acid under nitrogen protection at 120 °C for 8 h. The residual trifluoroacetic acid was removed by rotary evaporation. The hydrolysis product was added with ammonium hydrochloride (10 mg) and pyridine (1 mL) and reacted at 90 °C for 30 min. After cooling to room temperature, 1 mL of acetic anhydride was added and reacted at 90 °C for 30 min. The final product was extracted with 1 mL of chloroform and filtered through a 0.22 μm filter. The ensuing analysis was conducted by GC–MS (Polaris-Q, Thermo, USA) according to our previous study (Wei et al., 2019).

3. Results and discussion

3.1. Characterization of hydrogen bond interactions in DESs

As shown in Fig. 1a, the FT-IR spectra for HBAs, HBDs and the corresponding DESs were compared to prove hydrogen bond interactions. In DES_{1-4}, hydroxyl group stretching vibration was observed at 3333, 3347, 3354 and 3335 cm^{-1} respectively, which showed blue shifts compared with ChCl (3255 cm^{-1}). Similarly, it could be noted that asymmetric stretching vibrations of the methyl in TMAC were at 2914 cm^{-1}, which shifted to 2969 cm^{-1} for DES_5. On the other hand, the absorption peaks with strong intensity at 3357, 3374, and 3325 cm^{-1} were attributed to hydroxyl group stretching vibration of the polyol HBDs (EG, GLY and BDO), which exhibited absorption peaks shifts when DESs were formed. These findings indicated the intermolecular hydrogen bond interactions between HBA and HBD. It was supported that the addition of HBD weakened the intermolecular hydrogen bonds of ChCl itself, which caused the blue shift effect of hydroxyl group stretching vibration of ChCl, demonstrating the formation of hydrogen bonds between HBA and HBD (Wang et al., 2019). The study reported by Hou et al. also demonstrated that the forming of strong hydrogen bonds resulted in hydroxyl group stretching vibration of HBD shifting from 3331 to 3218 cm^{-1} (Hou et al., 2015).

The hydrogen bond interactions between HBA and HBD were further investigated via ^1H NMR spectra of DES_{1-8} (Fig. 1c). It was apparent that compared with HBA or HBD, the active hydrogen signal in corresponding DES upshifted for HBA and downshifted for HBD, which confirmed the existence of hydrogen bond interactions. As shown in Fig. S2, the chemical shift at 5.74 and 5.68 ppm corresponded to hydroxyl group of ChCl and hydrogen atoms of urea, which shifted to 5.52 and 5.87 ppm in ChCl-Urea, respectively. This result might be explained by the fact that the formation of intermolecular hydrogen bonds between ChCl and urea weakened the interactions between chloridion and hydroxyl group within ChCl, thus increasing the electron density of hydroxyl group and causing the chemical shift moving to high field. Meanwhile, hydroxyl group contributed to providing hydrogen bonding site resulting in the hydroxyl group signal of BDO shifting from 4.47 to 4.52 ppm in TMAC-BDO (Fig. S2). This result could be explained by the fact that the formation of hydrogen bonds caused the electron density of hydrogen atom in hydroxyl group reduction, which leading to the downshift of chemical shift (Hao et al., 2017). These findings matched the observation in previous studies that upshifting of the active hydrogen signal in HBA (tetraethylammonium hydroxide, from 0.91 to 0.74 ppm) and downshifting in HBD (GLY, from 3.681 to 3.798 ppm) (Sirviö & Heiskanen, 2020; Zaid et al., 2017). Overall, the results in this section indicated that hydrogen bonds formed between HBA and HBD.

3.2. Effect of hydrogen bond interactions on the dissolution of PAS in DESs

Based on the understanding of the hydrogen bonding properties of DESs, we wondered whether the abilities of DESs to solubilize heteropolysaccharide could be affected according to the modulation of HBAs/HBDs. First, the potential of dissolving PAS by the studied DESs was verified by the analysis of solubility and polarizing microscope. It could be seen in Fig. 2a that the solubility of PAS in DES was boosted up to 0.98 g/100 g compared to DMSO with much lower solubility (0.22 g/100 g). Moreover, what stood out in the data was that the solubilities of PAS in ChCl-based DESs improved compared to that in TMAC-based DESs in the case of having the same HBD. The polarity and Kamlet-Taft parameters of DESs were further investigated and the data from Table 1 suggested that the E_T^N and α values of ChCl-based DESs were

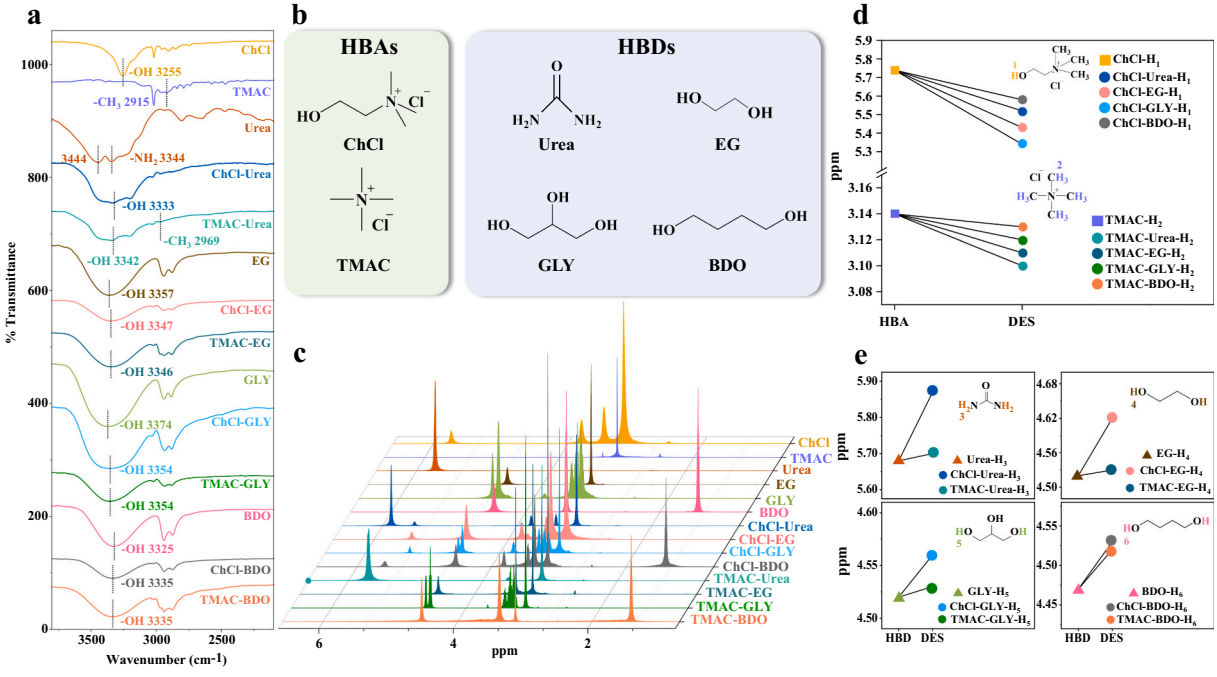


Fig. 1. Shifts of HBAs/HBDs spectra reflected hydrogen bonds: a) FT-IR and b) chemical structures and abbreviations of HBAs and HBDs for the preparation of DESs c) ^1H NMR spectra of DESs and corresponding HBAs and HBDs; comparison of ^1H NMR chemical shifts in DESs with d) HBAs and e) HBDs.

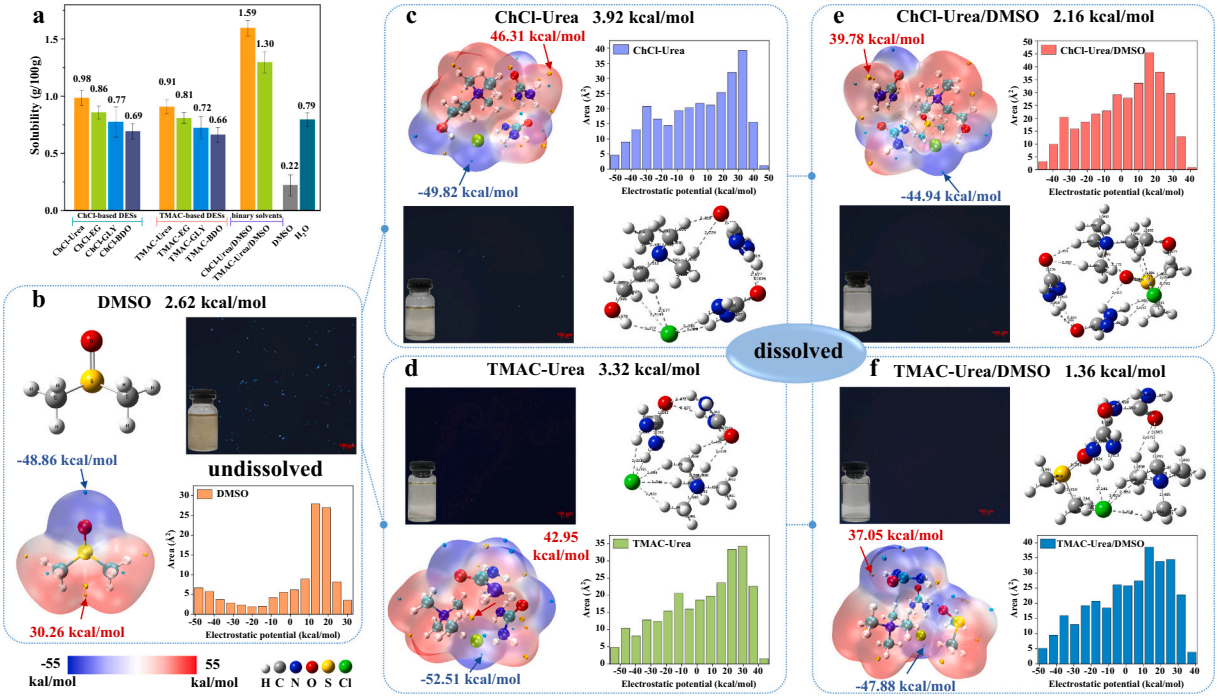


Fig. 2. a) Solubility of PAS in DESs and DES/DMSO binary solvents (DESs with the same HBA were marked with the same color); b) -f) each part consisted of the following contents: optimized geometry, polarizing microscope image of PAS dissolved in corresponding system, overall average value of ESP, molecular electrostatic potential values mapped on vdW surface and the percentage of area within each ESP range. The minimum and maximum values of the significant surface localization of ESP were denoted by cyan and yellow spheres.

larger than TMAC-based DESs with the same HBD, which seemed to be consistent with the solubilities. The high polarities of the ChCl-based DESs might be related to the existence of charged polar functional groups in the asymmetric chemical structure of ChCl (Florindo et al., 2018), which was beneficial to the dissociation of anions and cations, thus improving the activity of ions such as chloride ions (Ren et al., 2016). Meanwhile, the higher α values of ChCl-based DESs implied that the active hydroxyl groups in ChCl contributed to enhance the hydrogen bond donating ability of the whole system. These results suggested that ChCl-based DESs with suitable solvent properties were more conducive to interacting and disrupting the hydrogen bond networks of heteropolysaccharides, resulting in better dissolving capacity.

Further, an attempt was made to explore the driving forces and important influences behind the dissolution of PAS by DESs from the perspective of quantified hydrogen bond interactions through quantum chemistry calculations. The system structures and optimized conformations of ChCl-Urea, TMAC-Urea, ChCl-Urea/DMSO and TMAC-Urea/DMSO were shown in Fig. 2. It could be seen from the optimized geometries that the multiple active electronegative atoms and hydroxyl groups in the HBAs and HBDs structures conferred the ability to conduct hydrogen bonding interactions. And these atomic spacings were proven to be all within the acceptable criteria for hydrogen bond coordination (< 3.5 Å).

Molecular electrostatic potential was a valid means for analyzing and predicting hydrogen bond interactions and was applied here to analyze the contributing effect of PAS solubility in DES. Further, the positive and negative average values of ESP could quantitatively provide information related to the solubility of the condensed phase that depended on noncovalent interactions (Murray & Politzer, 2011). As shown in the data of ESP analysis, the solubility of PAS presented a certain positive relationship with the corresponding overall average ESP value of each DES. ChCl-Urea had the largest overall average value of ESP (3.92 kcal/mol), and the plot of its surface area in each ESP range showed that the maximum area range (14.31%) was positive 32.67. This might contribute to stronger interactions between ChCl-Urea and PAS with plenty of negative charge, thus improving the solubility of PAS. In addition, it could be seen from Fig. S4 that the overall average value of ESP in ChCl-BDO with lower dissolving capacity was only 0.86 kcal/mol, the surface area distribution in different ESP range was relatively uneven, with the largest area (25.04%) of just positive 7.00. Similarly, the overall lower average electrostatic potential of TMAC-Urea might influence the interactions with the hydroxyl groups of polysaccharides, resulting in lesser solubility.

The possible dissolution mechanism of PAS in ChCl-Urea was predicted by FMO analysis since the molecular orbitals were significant electronic parameters for describing the chemical reactivities of compounds. As shown in Fig. 3a, the highest occupied molecular orbitals (HOMO) electron cloud of ChCl-Urea was localized on chloridion, and the lowest unoccupied molecular orbitals (LUMO) electron cloud was mostly localized on the carbon and hydrogen atoms of the quaternary ammonium cation as well as the amino group of urea. These results could provide insights into forecasting the action sites between DES and PAS. It was well known that HOMO determined the region where electrons had the highest reaction energy, while LUMO established a region for filling with intruding electrons from other molecules during the bonding process (Dwamena & Raynie, 2020). Because of this, HOMO was related to the propensity to donate electrons, while LUMO was related to the propensity to accept electrons. Therefore, it was speculated that the LUMO region and HOMO region in ChCl-Urea might interact with the nucleophilic and potentiophilic sites in PAS, respectively. This finding was supported by the research of Zhang et al. that hydroxyl protons of cellulose could form strong hydrogen bonds with chloridions, and the intermolecular hydrogen bond networks of cellulose were broken during this process (Zhang et al., 2014).

According to the above solubility phenomena together with the theoretical calculations and analysis, the studied DESs were full of hydrogen bond interactions and the multiple hydrogen bonding sites within the system provided their abilities to interact with PAS. ChCl-based DESs became potential candidates for enhancing PAS solubility due to their tendency to have greater polarity and overall average ESP. It

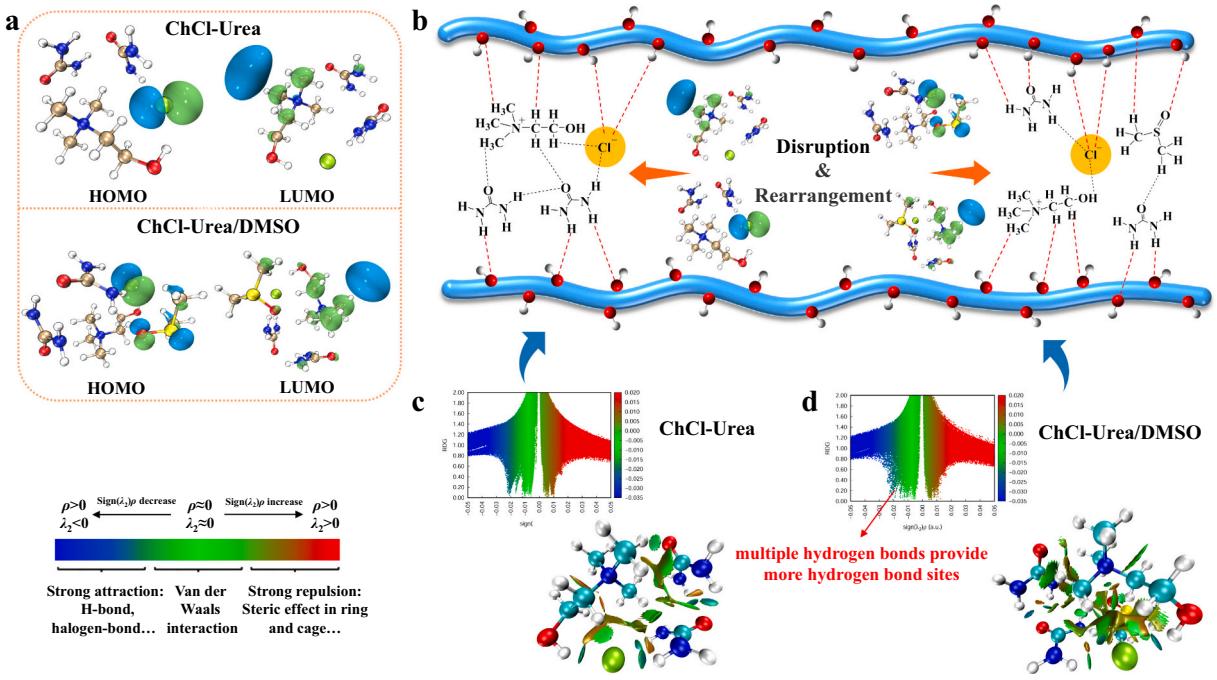


Fig. 3. a) The HOMO and LOMO orbitals of solvents; b) possible solubilization mechanism of polysaccharides in ChCl-Urea and ChCl-Urea/DMSO; c) -d) 2D/3D non-covalent interaction diagrams of ChCl-Urea and ChCl-Urea/DMSO. The large, negative values of sign(λ_2)ρ indicated hydrogen bond (blue). The large, positive values of sign(λ_2)ρ represented spatial repulsive (red), and the values close to zero were associated with vdW interactions (green).

was suggested that the structure of the HBA/HBD could be deployed to prepare the advantageous DES and thus improved the solubility of the heteropolysaccharide PAS.

3.3. Effect of co-solvent effect on PAS dissolution

In this work, considering the viscosity and cost of solvent in polysaccharide functionalization, the common organic solvent DMSO was chosen to construct binary mixed solvents system, and further exploring the potential of the DES/DMSO system. Surprisingly, from the perspective of solubility results (Fig. 2a), the maximum solubility of PAS in DES/DMSO was increased to 1.6 times as much as the corresponding pure DES. This also corresponded to the increased hydrogen bond basicity (β) of DES/DMSO system (Table S1), which promoted the interactions between mixed solvents and hydroxyl groups on polysaccharides, breaking the original polysaccharide hydrogen bond networks and creating new hydrogen bonds with polysaccharides (Tomimatsu et al., 2019). This was confirmed by related studies by Rogers et al. on the effect of solvent properties of ionic liquids for the solubilization of cellulose (Remsing et al., 2006; Wang, Gurau, & Rogers, 2012). Similarly, Mai et al. reported that the [Bmim][Cl]/DMSO mixture with higher β value exhibited significantly enhanced lignocellulose dissolving capacity (Mai et al., 2014). In addition, FT-IR was performed on regenerated PAS from binary mixed solvents, and there was no significant difference in the characteristic peaks between the untreated and regenerated PASs. There were no significant changes observed in the composition and molar ratio of monosaccharides in the regenerated polysaccharides compared to PAS (Fig. S5). Molecular mass analysis (Table 2) showed that the M_W of regenerated PASs from DES and DES/DMSO (PAS_{DES_1}, PAS_{DES_2}, and $PAS_{DES_2/DMSO}$) were observed to be slight reduced (4.86×10^4–5.98×10^4 g/mol). These findings indicated the chemical stability of regenerated PAS and no derivatization reaction between binary solvents and PAS.

Moreover, the binary mixed solvents were characterized in the same way as DESs to explore the function of co-solvent effect on solubility enhancement by comparing them. It could be observed from the optimized geometries of ChCl-Urea/DMSO and TMAC-Urea/DMSO (Fig. 2e and Fig. 2f) that the hydrogen bonds formed in the system were within the standard range. The RDG scatter plots and the colored RDG isosurfaces of ChCl-Urea and ChCl-Urea/DMSO in Fig. 3c and

Table 2
Se-functionalization of PAS facilitated by DES and DES/DMSO.

Sample	Solvent	Yield (%)	Selenium content (μg/g)	M_W (10^4 g/mol)	PD
PAS[b]	–[a]	–[a]	–[a]	6.96	2.16
PAS_{DES_1}[c]	DES_1	90.70	–[a]	5.98	2.68
PAS_{DES_2}[c]	DES_2	97.60	–[a]	5.49	4.09
$PAS_{DES_2/DMSO}$[c]	DES_2/DMSO	93.50	–[a]	4.86	3.06
$SePAS_C$[d]	0.7% HNO3	91.60	1703.00	1.14	1.78
$SePAS_{DMSO}$	DMSO	92.40	970.98	4.03	1.39
$SePAS_{DES_1}$	DES_1	99.20	1535.60	2.69	1.34
$SePAS_{DES_2}$	DES_2	92.50	2126.70	2.35	1.71
$SePAS_1$	DES_1/DMSO	94.00	1947.46	1.22	1.17
$SePAS_2$	DES_2/DMSO	61.00	4004.29	0.65	1.16
$SePAS_3$	DES_3/DMSO	89.00	2480.86	0.96	1.24
$SePAS_4$	DES_4/DMSO	90.20	1247.99	1.19	1.09
$SePAS_5$	DES_5/DMSO	92.00	2069.18	1.16	1.18
$SePAS_6$	DES_6/DMSO	68.00	2759.63	0.66	1.24
$SePAS_7$	DES_7/DMSO	91.40	2039.33	1.07	1.08
$SePAS_8$	DES_8/DMSO	89.70	1107.28	1.19	1.12

[a] Not detected.
[b] Data from our previous study (Wei et al., 2019).
[c] No selenylation reagent added.
[d] $SePAS_C$ was prepared by conventional HNO_3-H_2SeO_3 method and data from our previous study (Wang, Zhao, et al., 2012).

Fig. 3d further quantified more types of non-covalent interactions. It was revealed that the introduction of DMSO gave rise to more spikes located at low density and low gradient in the negative region compared to its corresponding DES, while the distribution range was broadened. This indicated the existence of more complex hydrogen bond networks within the binary mixed system, which facilitated the hydrogen bonds synergy during the dissolution process (Kalhor et al., 2020). Furthermore, the hydrogen bonds were investigated quantitatively by integrating real space functions within RDG isosurface. The results of the electron density integration showed that the number of electrons involved in the merged hydrogen bond domains of ChCl-Urea/DMSO system (0.0547 au) was larger than those of ChCl-Urea system (0.0424 au), which was closely related to the strength of the interaction (De Silva & Corminboeuf, 2014). Meanwhile, the hydrogen bond domains in ChCl-Urea/DMSO system (3.283 $Bohr^3$) involved wider spatial region than those in ChCl-Urea system (2.360 $Bohr^3$). These results further demonstrated more and stronger hydrogen bonds within ChCl-Urea/DMSO system.

In the case of ESP analysis, the maximum value ($V_{S,\ max}$) and the minimum value ($V_{S,\ min}$) exhibited correlation with the hydrogen bond providing ability and hydrogen bond accepting ability, respectively (Hagelin et al., 1995). Fig. 2 provided the ESP local extremums of ChCl-Urea, DMSO and ChCl-Urea/DMSO. The $V_{S,\ max}$ (46.31 kcal/mol) of ChCl-Urea was near a hydrogen atom that belonged to one of the urea, and this region was more likely to interact with the oxygen atom that presented the surface minimum of ESP in DMSO (−48.86 kcal/mol). This also accorded with the ESP mapped molecular vdW surface of ChCl-Urea/DMSO and above optimized structure. The overall value of ESP for ChCl-Urea/DMSO (2.16 kcal/mol) was decreased compared to that of ChCl-Urea, presumably due to the effect of the highly negative region of DMSO on the overall ESP of the binary mixed system. Clark et al. reported that the overlap of oxygen atom and sulfur atom in DMSO to form extremely negative regions (−46.0 kcal/mol), which allowed DMSO to readily interact electrostatically with other molecules (Clark et al., 2008). Besides, the FMO analysis further provided relevant insights into the action sites between the binary mixed system and PAS. It was observed in Fig. 3a that compared to ChCl-Urea, chloridion and DMSO together constituted the HOMO electron cloud localization region for ChCl-Urea/DMSO. The LUMO electron cloud of ChCl-Urea/DMSO was increased compared to ChCl-urea and appeared mainly near the methyl group of ChCl, the amino group and oxygen atom of urea, and the oxygen atom and methyl group of DMSO. Hence, it could be hypothesized conceivably that the combination of DMSO and DES provided more sites for the binary mixed system to interact with PAS, resulting in stronger solubilization ability.

The AIM theory was used to investigate the critical points between the HBD and HBA nuclei as well as measure the hydrogen bond interactions in terms of the electron density (ρ) and the Laplacian of the electron density ($\nabla^2\rho_{BCP}$) at the bond critical point (BCP). As shown in Table S2, the values of $\nabla^2\rho_{BCP}$ were observed as all positive, proving that they were all within the range of standard hydrogen bond (0.024–0.139 au) for ChCl-Urea, TMAC-Urea and their corresponding binary mixed solvents. By comparing the ρ values associated with the bond strength of DES and DES/DMSO, it could be seen that part of the hydrogen bonds in DES system were weaken by the addition of DMSO. This contributed to the dissociation of the anions and cations in DES, allowing the ions and HBD to achieve easier interactions with PAS, thus facilitating solubilization. Likewise, Zhao et al. held the view that DMSO partially disrupted the ionic binding in [C_4mim][CH_3COO] and promoted the binding of [CH_3COO]$^-$ to cellulose, resulting in improved solubility of cellulose in [C_4mim][CH_3COO]/DMSO binary solvents (Zhao et al., 2013). On the other hand, there was an increase in the number of hydrogen bonds within binary system and an improvement in the strength of the strongest hydrogen bond ($O_{32}\cdots H_{30}-N_{25}$, 0.0236 au, ChCl − Urea/DMSO) compared to ChCl-Urea ($O_{32}\cdots H_{30}-N_{25}$, 0.0222 au), which provided the binary mixed solvents with more and stronger hydrogen bonding sites for interacting with PAS and facilitating its superior solubility.

To sum up the above analysis, the DES/DMSO system could break the hydrogen bond networks of PAS even better because possessed more and stronger hydrogen bond interactions compared to the corresponding DES. The addition of DMSO disrupted the original hydrogen bond networks of DES and reduced the viscosity of the solvent system, which allowed the components within the system to interact more easily with polysaccharides, so that the DES/DMSO system had the capability to act as a medium for PAS solubilization and further functionalization reactions. However, the multiple hydrogen bonds between polysaccharide chains were the source of the aggregation and insolubility of polysaccharides, and there were still many factors (branching structure, hydrophobic groups, conformation and solvent characteristic) affecting this complex process of polysaccharide dissolution (Guo et al., 2017). Therefore, it was necessary to conduct comprehensive investigations on the solubilization factors of polysaccharide solvents in the future.

3.4. Se-functionalization of PAS promoted by hydrogen bond modulation

In this work, with the purpose of constructing a novel polysaccharide functionalization system, PAS was firstly Se-functionalized in the reaction system with DES as the solvent, while related control experiments were carried out under different solvent conditions at the same time. Table 2 presented selenium contents and yields of SePASs obtained under different conditions. Due to the weak solubility of PAS in DMSO (Fig. 2a), the selenium content of $SePAS_{DMSO}$ (970.98 μg/g) obtained by the selenylation reaction conducted in DMSO were significantly lower. When DESs as reaction media, the selenium contents of corresponding $SePAS_{DES_1}$ and $SePAS_{DES_2}$ were elevated to 1535.60 and 2126.70 μg/g, respectively. It indicated that adequate solubilization of polysaccharide had a significant impact on determining the efficiency of subsequent functionalization. DES exhibited the dual properties of hydrogen bond providing and accepting abilities, and this characteristic brought about the superior solubilizing capacity of DES, which was supported by the above experimental and computational analysis. Therefore, it was probable that DES could break the hydrogen bond interactions between polysaccharide chains and expose more hydroxyl groups. In addition, the high viscosity of ChCl-Urea impeded the contact of substrates in the reaction system under sufficient solubilization condition, resulting in a compromised selenylation efficiency of $SePAS_{DES_1}$. The excellent dissolving capacity and lower viscosity of ChCl-EG might cause the selenogroups to be more easily introduced into the polysaccharide chains, thus enhancing the efficiency of the selenylation reaction. Consequently, DES with multiple hydrogen bonding sites was capable of being a new type of selenylation medium.

On the basis of the promising Se-functionalization effect of DES system, the co-solvent effect on the selenylation efficiency of PAS in DES/DMSO binary mixed solvents system was further explored. The selenium contents of $SePAS_1$ and $SePAS_2$ were 1947.46 and 4004.29 μg/g, which increased to a certain extent compared with the corresponding $SePAS_{DES_1}$ (1535.60 μg/g) and $SePAS_{DES_2}$ (2126.70 μg/g) obtained in pure DES systems. As shown in Fig. S6, the co-solvent strategy could reduce the viscosity of the reaction system and facilitate the contact between polysaccharides and H_2SeO_3 (Damstrup et al., 2006). Additionally, the binary mixed solvents system was shown in theoretical calculations and analysis to provide more hydrogen bonding sites in interaction with PAS (Section 3.3), allowing adequate exposure of PAS hydroxyl groups. The combination of the above two factors could lead to a better nucleophilic attack by hydroxyl groups, thus favoring the polysaccharide selenylation reaction. Similarly, Harifi-Mood et al. studied the reaction of 1-fluoro-2,4-dinitrobenzene with piperidine in DES/DMSO binary mixed solvents and found that the increase of DMSO was beneficial to elevate the reaction rate (Harifi-Mood & Sadrzadeh, 2018). Moreover, selenium contents of $SePAS_{1-8}$ were higher than that of $SePAS_C$ (1703.00 μg/g) obtained by conventional system (HNO_3–H_2SeO_3) (Wang, Zhao, et al., 2012). This valuable result

indicated that DES/DMSO binary mixed solvents system overcame the drawback of the conventional aqueous solvent with lower esterification reaction efficiency. On the other hand, the good solubilization ability of DES/DMSO extended the range of substrates and improved the homogeneity of the reaction system (Fig. 4). In short, DES/DMSO binary solvents strategy was an applicable approach for the efficient preparation of selenium polysaccharide.

Moreover, a certain positive correlation between the selenium contents of SePASs and the polarities of DES/DMSO systems was confirmed by the experimental results. The higher polarities of ChCl-based DESs/DMSO compared to TMAC-based DESs/DMSO also corresponded to the higher selenium contents of the selenium polysaccharides obtained from ChCl-based DESs/DMSO. The high polarity of solvent facilitated the dissociation of chloridion, which was more favorable for hydroxyl groups exposure and subsequent selenylation reaction. This was in keeping with a reported work, which reported that the highly polar solvents could promote the esterification reaction of lactic acid and propyl glycoside by increasing the solubility of the substrate (Wei et al., 2003). In summary, by reasonably combining and fully exploiting the synergistic effect of HBA with HBD on the solvent property, it was possible to achieve prospective regulation of Se-functionalization efficiency.

3.5. FT-IR and 1D/2D NMR analysis of SePAS

FT-IR spectra of PAS and its selenized derivatives ($SePAS_2$ and $SePAS_6$) were presented in Fig. 5a. The characteristic peaks at 3395 and 2926 cm^{-1} corresponding to O—H and C—H stretching vibrations of polysaccharide. The most significant result was that there were two new peaks appeared at 1076 and 807 cm^{-1} in SePAS compared with PAS, which were attributed to Se═O stretching vibration and C-O-Se bending vibration, respectively. This finding was consistent with the study of Feng et al. that two characteristic absorbance bands at 1130 and 742 cm^{-1} were assigned to Se═O asymmetric stretching and C-O-Se vibration in selenized *Chuanminshen violaceum* polysaccharide (Feng et al., 2016).

NMR was adopted to obtain a deeper insight into the structure of polysaccharide and seleno-group substitution position (Fig. 5). Our previous study demonstrated the characteristic chemical shifts of PAS (Wang et al., 2015), which was consistent with the results obtained in this study. For $SePAS_2$ and $SePAS_3$, it was apparent that new carbon signal of β-D-Man C-6 was observed at 68.37 ppm when compared with PAS. Meanwhile, combined with COSY and HSQC spectra (Fig. 5c and Fig. 5d) and data from references, it could be concluded that new proton and carbon signals about C-6 of β-D-Man appeared at 3.38/68.37 ppm, and the chemical shift of the H-5 (3.37 ppm) adjacent to H-6 (3.38 ppm) was also been further confirmed (Wang et al., 2018; Zhang et al., 2021). Overall, these results suggested that O-6 position of β-D-Man occurred substitution reaction, which could lead to the introduction of seleno-groups. This finding was supported by Zhu et al. who held the view that the main substitution position of synthetic selenized green tea polysaccharide (CSe-tps1) was the C-6 position (71.82 ppm) (Zhu et al., 2020). Additionally, for $SePAS_2$, the C-6 peaks of Gal (4.05/62.68 ppm), Glc (3.68/61.18 ppm), and Man (3.38/59.82 ppm) still existed, which demonstrated that the hydroxyl groups on C-6 were not completely replaced. Likewise, Wei et al. pointed out that the C-6 peaks in selenium modified *Radix hedysari* polysaccharide (Se-RHP) were not completely disappeared, which suggested that the C-6 substitution was incomplete (Wei et al., 2015). From the above discussion, it could be seen that PAS was incompletely replaced through selenylation reaction, and the substitution reaction occurred at the C-6 position.

3.6. Polysaccharide chain features following DES/DMSO-mediated Se-functionalization

Through the analysis of molecular mass and monosaccharide composition, the polysaccharide chain characteristics of SePASs were further explored. As shown in Table 2, the M_W and polydispersities (PD) of SePASs prepared in various pure DESs and DES/DMSO binary mixed solvents were obtained by SEC-MALLS. In conventional method of polysaccharide selenylation reaction, the M_W of $SePAS_C$ decreased significantly to 1.14×10^4 g/mol due to the strongly acidic environment provided by $HNO_3 - H_2SeO_3$ (Wang, Zhao, et al., 2012). As indicated in Section 3.3, the dissolution of PAS in DES and DES/DMSO had little effect on the degradation of the polysaccharide chains, which was reflected in M_W data of PAS_{DES_1}, PAS_{DES_2} and $PAS_{DES_2/DMSO}$ (Table 2). After

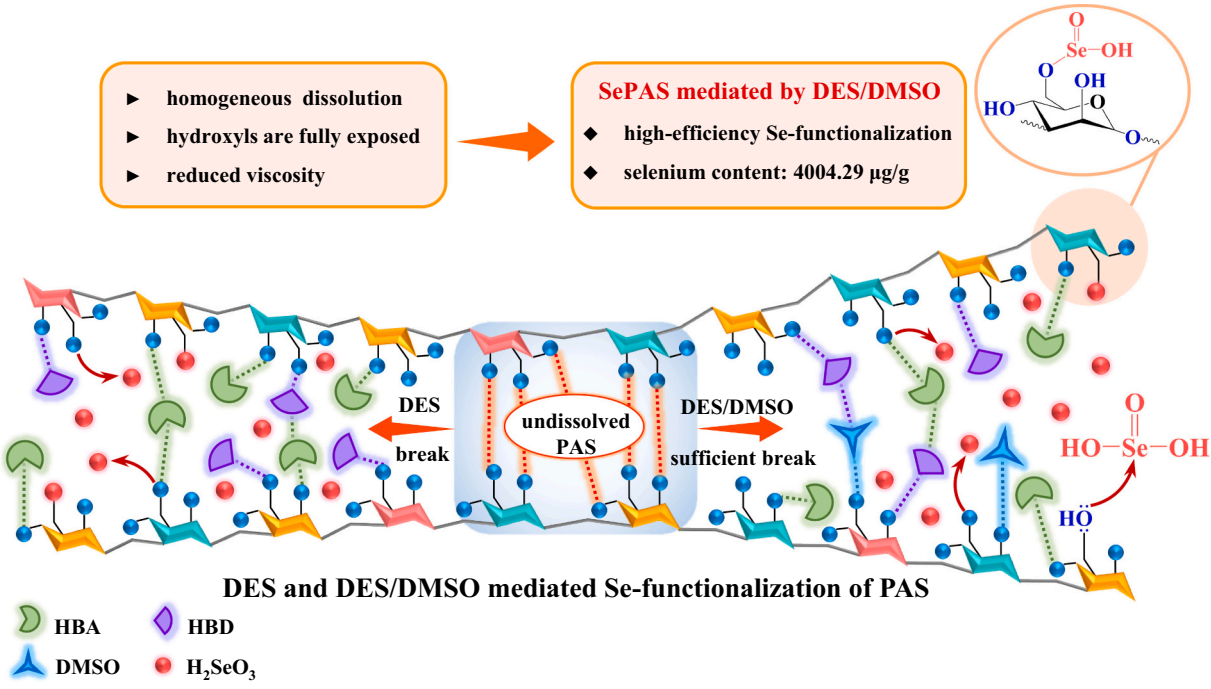


Fig. 4. Schematic diagram of DES and DES/DMSO-mediated PAS Se-functionalization.

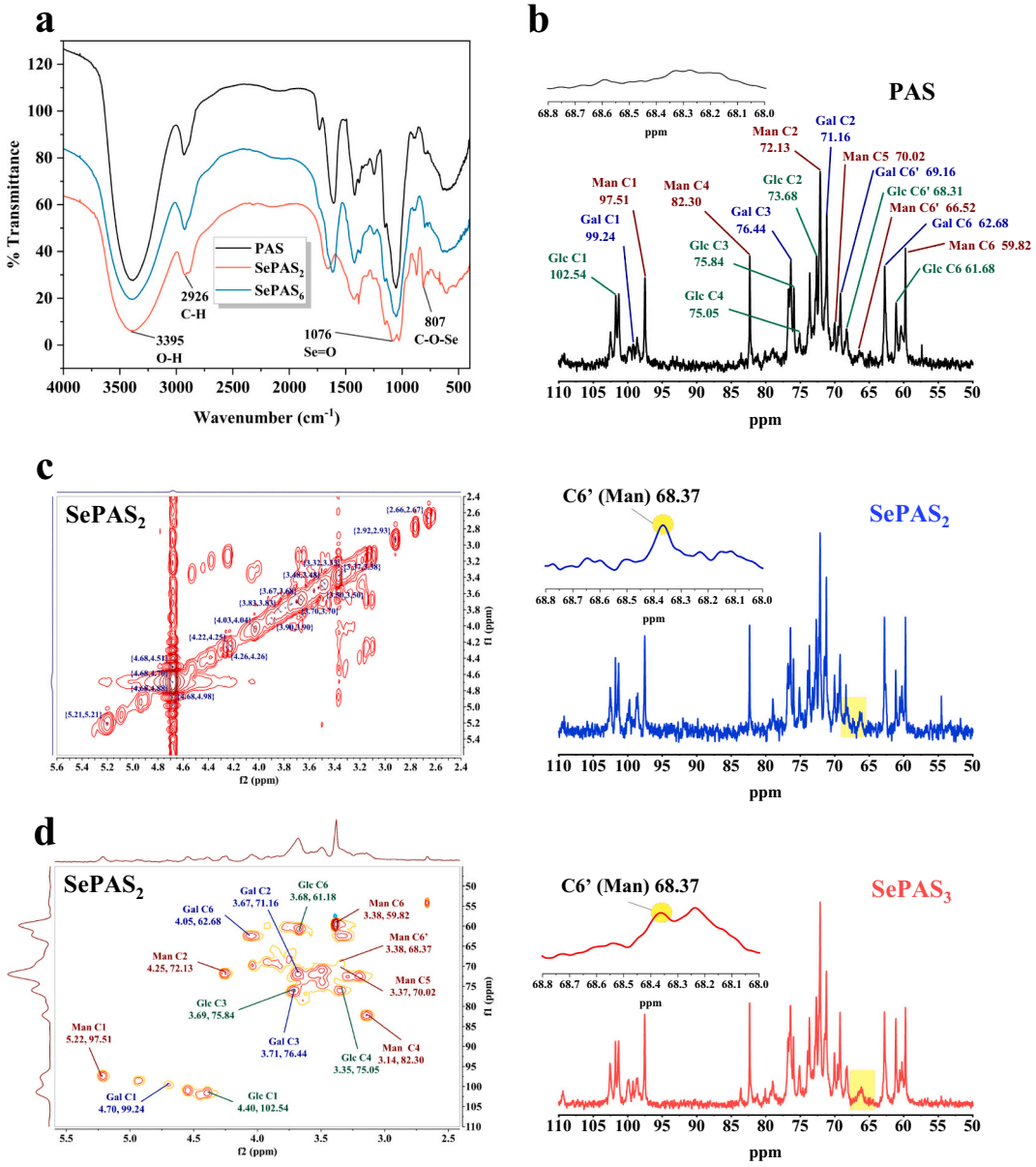


Fig. 5. a) FT-IR spectra of PAS, $SePAS_2$, and $SePAS_3$; b) ^{13}C NMR spectra of PAS, $SePAS_2$ and $SePAS_3$; c)- d) COSY and HSQC spectra of $SePAS_2$.

Se-functionalization mediated by H_2SeO_3, M_W of $SePAS_{DES_1}$ and $SePAS_{DES_2}$ reducing to 2.69 and 2.35 $\times$ 10^4 g/mol, suggesting that acidic environment was an important driving actor affecting the M_W of polysaccharide derivatives.

Additionally, under the condition that DES and DMSO formed binary mixed system, the M_W of $SePAS_1$ and $SePAS_2$ decreased to 1.22 and 0.65 $\times$ 10^4 g/mol compared with that of $SePAS_{DES_1}$ and $SePAS_{DES_2}$ (pure DES_1 and DES_2 as solvent), which could be attributed to co-solvent effect. For one thing, DMSO interacted with DES after it was added, further reducing the DES viscosity (Fig. S6). This change of solvent property contributed to the adequate contacts and interactions of the substances in the reaction system. For another thing, the above AIM calculative analysis suggested that DMSO addition destroyed the original interactions within DES itself, which might lead to the dispersion of DES aggregates and promote the components of DES to interact with polysaccharides and to destroy hydrogen bond networks structure between or within polysaccharides chains, thus facilitating the conversion of SePAS to lower M_W. This was in agreement with a recent study that the addition of DMSO could result in the dissociation of [Bpy][DCA] ionic liquid from large ion clusters into small ion pairs or even into isolated ions (Zhang et al., 2018). Meanwhile, the finding of M_W reduction of polysaccharide derivatives was supported by the result from monosaccharide composition analysis (Fig. 6b). It was found that the major components of $SePAS_1$ and $SePAS_2$ remained unvaried after selenylation, all of which were Ara, Xyl, Man, Glc and Gal. The ratio of Gal/Man (0.80) in $SePAS_1$ and $SePAS_2$ decreased to 0.79 and 0.55, and the ratio of Gal/Glc (0.56) decreased to 0.52 and 0.36, respectively. It might be related to the degradation of the polysaccharide chain leading to the loss of low M_W polysaccharides or oligosaccharide fragments during dialysis (Wang et al., 2017). Taken together, it could be suggested that acidic environment was a major influence contributing to the alteration of the M_W of polysaccharide, and the co-solvent effect within DES and DMSO cooperated with the acidic environment in selenylation process to become a driving force for the degradation of polysaccharides into fragments with different M_W.

It was found that under the same HBA condition in binary system, the different degradation degrees of polysaccharides were affected by the polarity parameters of DES/DMSO systems. Basically, the M_W of

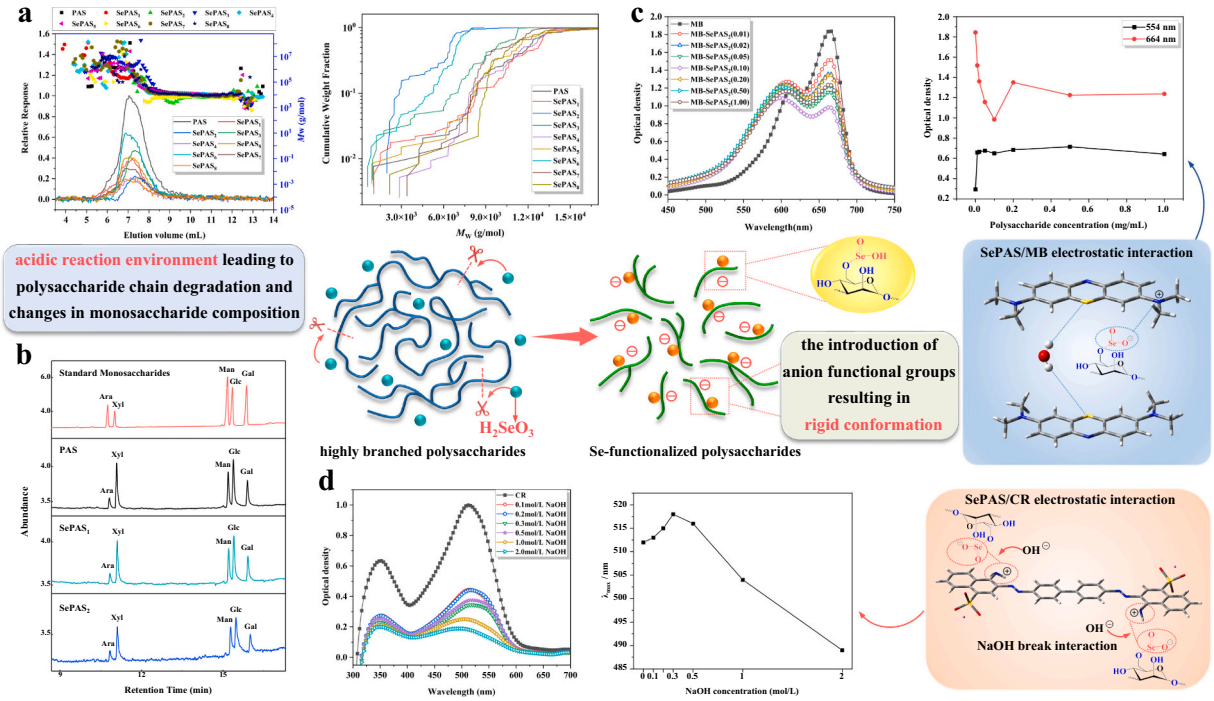


Fig. 6. a) SEC-MALLS chromatograms of PAS and $SePAS_{1-8}$: laser light scattering photometry and molecular mass distributions; b) monosaccharide composition analysis of standard monosaccharides, PAS, $SePAS_1$ and $SePAS_2$; c) absorption spectra of MB measured from 450 nm to 750 nm at different concentrations of $SePAS_2$ at room temperature and evolution of optical density values at 664 nm and 554 nm for $SePAS_2$ as a function of polysaccharide concentration; d) absorption spectra and evolution of maximum absorption wavelength for CR-$SePAS_2$ (0.2 mg/mL) in various concentrations (mol/L) of NaOH.

$SePAS_{1-8}$ had a certain correlation with the polarity of the corresponding DES/DMSO, and the overall rule presented a trend that the higher polarity (higher E_T^N value) of DES/DMSO, the lower M_W of selenium polysaccharide. This relationship was likely to be related to the fact that high polarity solvent was more conducive to the dissociation of anions and cations, which was beneficial to the destruction of hydrogen bonds within polysaccharide chains by chloridion and the subsequent major degradation of polysaccharide chains by acidic reaction environment. This also corresponded to the finding that both solubility and selenium content showed a correlation with polarity of DES. Likewise, the study of Liu et al. reported that the hydrogen bonds between chloride ions of DES and hydroxyl groups might weaken the strong hydrogen bond interactions in wood lignocellulose, thus breaking lignin-carbohydrate complexes linkages (Liu et al., 2017).

Furthermore, the evolution of chain conformation in polysaccharide Se-functionalization mediated by DES/DMSO was investigated using MB and CR spectrophotometric method (Prentø, 2001). With increasing $SePAS_2$ concentration, the gradual decrease at OD_{664nm} of MB/$SePAS_2$ absorption spectra and the progressive increase at OD_{554nm} indicated that the positively charged MB interacted with the negatively charged $SePAS_2$, which proved that seleno-groups were successfully introduced (Fig. 6c). Following this interactions, a long-range forces involving water and MB molecules appeared to bring the planar cationic molecules relatively close to each other and form metachromatic complexes, leading to selenium polysaccharide more inclined to rigid conformation (Michon et al., 2002). Meanwhile, it was observed that when $SePAS_2$ concentration was 0.2 mg/mL, a large increase in NaOH destroyed the $SePAS_2$/CR electrostatic interactions and disrupted the tendency for λ_{max} of $SePAS_2$/CR to be continuously red-shifting (Fig. 6d), which suggested that the rigid conformation of selenium polysaccharide (Tan et al., 2020). Similar results were reported by Liu et al. that the upward trend in λ_{max} of polysaccharide/CR was influenced by the increasing NaOH concentration, confirming the extended conformation of polysaccharide (Liu et al., 2016).

According to the above results, it was found that the M_W of SePAS was related to the polarity of DES, owing to the positive contribution of the high polarity DES to destruct the hydrogen bond networks of polysaccharides, which further affected the degradation of polysaccharides by acid environment. Additionally, the DES/DMSO-mediated introduction of seleno-group rendered the polysaccharide chains towards more rigid conformation. In summary, it might be possible to prospectively modulate the selenylation efficiency and polysaccharide M_W simultaneously by constructing task-specific DESs with different HBAs/HBDs and solvent properties, which also provided new perspectives for the selection of solvent systems for polysaccharide derivatization reactions.

4. Conclusion

The DES/DMSO binary solvents with more and stronger hydrogen bonding sites could coordinate the disruption and reconstruction of hydrogen bonds within polysaccharides, attaining superior solubilization ability than DES. The polysaccharide hydroxyl groups were fully exposed and activated in DES/DMSO-mediated novel Se-functionalization system owing to the disruption of the hydrogen bond networks, which facilitated selenylation efficiency and rendered acid-dependent M_W reduction. The obtained SePAS presented a tendency of rigid conformation due to the introduction of anion functional groups. In summary, this study supported further development of DESs with regulable solvent properties towards solubilization and functionalization high-value exploitation of heteropolysaccharide bioresources.

Abbreviation

DES	deep eutectic solvent
DMSO	dimethyl sulfoxide
PAS	*Artemisia sphaerocephala* polysaccharide
SePAS	Se-functionalized *Artemisia sphaerocephala* polysaccharide
HBA	hydrogen bond acceptor

HBD	hydrogen bond donor
ChCl	choline chloride
TMAC	tetramethylammonium chloride
EG	ethylene glycol
GLY	glycerol
BDO	1,4-butanediol
FT-IR	fourier transform infrared
NMR	nuclear magnetic resonance
HSQC	heteronuclear singular quantum correlation
COSY	homonuclear shift correlated spectroscopy
HNO_3	nitric acid
H_2SeO_3	selenious acid
M_W	weight average molecular mass
SEC-MALLS	size-exclusion chromatographic analysis with multiangle laser light scattering detection
GC–MS	gas chromatography-mass spectrometer
MB	methylene blue
CR	congo red

CRediT authorship contribution statement

Yaxu Sun: Data curation, Writing – original draft. **Xiaoyan Jia:** Visualization, Investigation. **Ru Yang:** Visualization, Investigation. **Xiaojie Qin:** Data curation, Visualization, Investigation. **Xiaoxue Zhou:** Visualization, Investigation. **Hui Zhang:** Visualization. **Weibao Kong:** Visualization. **Ji Zhang:** Visualization. **Junlong Wang:** Supervision, Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbpol.2022.119159.

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